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Paleomagnetism of the Middle Proterozoic Electra Lake Gabbro, Needle Mountains, southwestern Colorado

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Abstract. The Electra Lake Gabbro is a small 1.435 Ga pluton that intrudes 1.7 to 1.6 Ga gneisses and schists of the Needle Mountains in southwestern Colorado. Paleomagnetic samples were collected from the main phases of the gabbro, diabase dikes, granite, and alaskite dikes that cut the gabbro and from a partially melted zone in gneiss along the southern margin of the pluton. Gabbro, diabase, and some melt zone samples have a single-polarity characteristic magnetization of northeast declination (D) and moderate negative inclination (I). Demagnetization behavior and rock magnetic characteristics indicate that the remanence is carried by nearly pure magnetite. After correction for the minor west dip of overlying Paleozoic strata, we obtain a mean direction of $D = 32.1°$, $I = -41.9°$ ($k = 94$, $\alpha_{95} = 3.3°$, $N = 21$ sites) and a paleomagnetic pole at 21.1°S, 221.1°E, ($K = 89$, $A_{95} = 3.4°$). This pole is similar to poles from the Middle Proterozoic Belt Supergroup but is located at a higher southerly latitude than poles from other 1.47-1.44 Ga plutons from North America, most of which plot at equatorial latitudes. The reason for this discrepancy is not clear but may result from a combination of factors, including unrecognized tilting of the gabbro, the failure of this relatively small pluton to fully average paleosecular variation, and uncertainties in the overall reliability of other 1.5-1.4 Ga poles of the North American apparent polar wander path.

1. Introduction

The circa 1.5-1.4 Ga part of the North American Middle Proterozoic apparent polar wander (APW) path is considered to be fairly well-defined [*Harlan et al.*, 1994]. Paleomagnetic poles of this age are derived primarily from Middle Proterozoic intrusions exposed from Wyoming to Labrador or from sedimentary strata of the Belt Supergroup of western Montana. The poles from the circa 1.47-1.43 Ga intrusions are reasonably well clustered and are located in the central Pacific ocean at equatorial latitudes, whereas those from the Belt Supergroup have latitudes of 10° to 25°S. The angular distance between a mean pole for the Belt Supergroup and the 1.47-1.43 Ga intrusions is 22° and is statistically significant. The apparent discordance between the two groups of poles has been attributed to a combination of apparent polar wander (APW), possibly due to plate motion, and uncertainties regarding the age of remanence acquisition in some of the rocks from which the poles are derived [*Harlan et al.*, 1994]. The discordance between the two groups is striking, given that recent U-Pb dating of rocks from the Belt Supergroup demonstrates that most of the rocks from which the two groups of poles are derived are approximately coeval [*Aleinikoff et al.*, 1996]. Although uncertainties exist regarding the age of remanence acquisition in some of these rocks, the disparity between the two groups suggests that important problems exist with respect to the reliability of some or all of the poles that comprise the circa 1.5-1.4 Ga part of the North American APW path.

A better understanding of the discordance between paleomagnetic poles from the Middle Proterozoic sedimentary and igneous rocks from North America requires further high-quality, well-dated paleomagnetic results. Here we present new paleomagnetic and $^{40}Ar/^{39}Ar$ geochronologic results from the Electra Lake Gabbro (ELG) of southwestern Colorado. The ELG is an important locality because it is nonconformably overlain by Middle Cambrian sandstone, making it one of the few Proterozoic intrusions in western North America for which the effects of Phanerozoic tilting can be corrected. It is also one of only two Middle Proterozoic mafic complexes exposed in the southern Rocky Mountains. The location of the ELG pole is significantly south of poles from intrusive rocks elsewhere in North America and generally east of most poles of the Belt Supergroup. Hence, although this pole alone cannot be used to resolve the discrepancy between the other circa 1.5-1.4 Ga intrusive and sedimentary poles, the overall consistency of this pole with other Middle Proterozoic poles demonstrates coherence of this part of western North America with the Laurentian craton since about 1.4 Ga.

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2. Geology and Paleomagnetic Sampling

The Needle Mountains and adjacent ranges of southwestern Colorado expose about 1100 km² of Early to Middle

Proterozoic rocks (Figure 1). Early Proterozoic rocks were deformed and metamorphosed circa 1.75-1.65 Ga during Early Proterozoic accretion along the southern margin of the Laurentian craton. The Early Proterozoic basement rocks were intruded by voluminous Middle Proterozoic plutons about 1.4 Ga. In the southern Needle Mountains, *Gonzales et al.* [1994] divided the Proterozoic basement rocks into four groups: (1) an Early Proterozoic granite-greenstone complex (Irving Formation, Twilight Gneiss) with apparent ages of about 1.80-1.75 Ga; (2) synorogenic or postorogenic circa 1.73 and 1.70 Ma granitoids of the Tenmile Granite and Bakers Bridge Granite; (3) a thick package of deformed, greenschist facies fluvial and marine siliciclastic rocks of the Vallecito Conglomerate and the Uncompaghre Formation that locally overlie, or have been faulted against, the older rocks; and (4) Middle Proterozoic calc-alkaline granitic to gabbroic intrusive rocks, including the Eolus granite, ELG, quartz diorite of Pine River and the Trimble Granite. Detailed descriptions of the Early Proterozoic rocks of the Needle Mountains are provided by *Gonzales et al.* [1994].

The ELG is a relatively small (16 km^2) plutonic complex (Figure 2) exposed along the western margin of the Needle Mountains. It is roughly elongate in plan view, with a length of 4.6 km in a northwest direction and 2.6 km in the northeast direction. The geologic relations and petrology of the ELG are described by *Cross and Hole* [1910], *Barker* [1969], *Ghuma* [1971], *Tweto* [1987], and *Gonzales et al.* [1994, 1996]. The descriptions of the ELG presented below are based primarily on *Ghuma* [1971], *Gonzales et al.* [1994], and our own observations. The ELG is a medium- to coarse-grained ophitic gabbro with minor diorite and granodiorite. Hypersthene, augite, and plagioclase are the primary igneous minerals in most of the complex, but hornblende and biotite are found as both primary and deuteric phases. Quartz is common in more differentiated parts of the complex, where it comprises as much as 10-15% of the rock [*Ghuma*, 1971]. The gabbro is cut by irregular masses and dikes of pegmatitic gabbro composed primarily of biotite and plagioclase and by pink alaskite dikes and k-spar megacrystic granite dikes that may be related to the Eolus Granite.

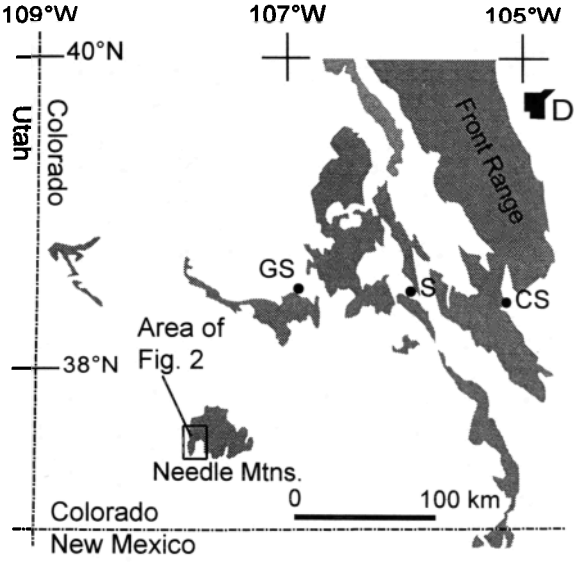


Figure 1. Sketch map showing the distribution of Precambrian basement rocks exposed in Colorado and the location of the study area in the Needle Mountains and the area of Figure 2. D is Denver, GS is Gunnison, S is Salida, and CS is Colorado Springs

Overall, the gabbro is massive and equigranular but locally shows a well-defined layering that strikes E-W to N40°W and dips 75°N to 90°N [*Ghuma*, 1971; *Gonzales et al.*, 1994]. The layering is defined primarily by megascopic compositional differences and the tabular alignment of plagioclase crystals and/or clinopyroxenes. *Ghuma* [1971] interpreted the layering as due to the effects of crystal settling. He suggested that the base of the complex, defined by cumulate orthopyroxenes and pronounced layering, is located along the northern contact and that the gabbro is increasingly differentiated toward the south. Mineralogic and compositional differences in the gabbro resulted from fractional crystallization of the magma. Because of the steeply dipping layering, *Ghuma* [1971] suggested that the complex was tilted about 90° from its original configuration. Alternatively, *Gonzales et al.* [1994] interpret the layering as a magmatic foliation limited to the margins of the complex. Our observations indicate that this foliation is present throughout much of the gabbro.

The emplacement age of the ELG is firmly established by several isotopic methods. Zircon from a pegmatitic gabbro gave an age of 1435 ± 2 Ma [*Gonzales et al.*, 1994] that we take as the best estimate for the age of the pluton. Sm-Nd whole rock and mineral isochrons gave similar ages of 1430 ± 25 Ma and 1451 ± 41 Ma [*Gonzales et al.*, 1994]. A whole rock Rb-Sr isochron gave 1425 ± 50 Ma [*Bickford et al.*, 1969]. U-Pb age determinations from the Eolus Granite gave essentially identical values of 1438 ± 20 Ma, 1435 ± 7 Ma, 1438 ± 9 Ma, and 1442 ± 3 Ma (*Silver and Barker* [1968] and unpublished data of D.A. Gonzales cited by *Gonzales et al.* [1996]). An alaskite dike that cuts the gabbro [*Silver and Barker*, 1968] gave a U-Pb date of 1438 ± 20 Ma. A whole rock Rb-Sr isochron from the granite gave an age of 1447 ± 27 Ma [*Bickford et al.*, 1969]. Crosscutting relationships show that the eastern part of the ELG cuts the Eolus Granite, but the abundant alaskite dikes cutting the gabbro and the similarity between the U-Pb dates indicate that the intrusions are coeval.

The Middle Proterozoic ELG and Eolus Granite have experienced a relatively simple postemplacement history as suggested by a single biotite $^{40}Ar/^{39}Ar$ analysis from the ELG. The biotite gives a simple, easily interpreted age spectrum (Figure 3 and Table 1), with only minor apparent ^{40}Ar loss shown by low-temperatures steps. A weighted mean calculated from the 1025°-1500°C steps gives a plateau date of 1431 ± 3 Ma (2σ) (Figure 3). This value is analytically indistinguishable from the ELG U-Pb zircon date. The combined U-Pb and $^{40}Ar/^{39}Ar$ geochronologic data indicate that the ELG cooled rapidly to temperatures below 300° ± 50°C (the closure temperature of biotite to ^{40}Ar diffusion). The relatively simple age spectrum indicates that the ELG has not experienced significant disturbance since closure to ^{40}Ar diffusion at about 1431 Ma.

Structural and kinematic analysis of the contact zone between the ELG and the Early Proterozoic gneisses and amphibolites established that the ELG and Eolus Granite were synkinematically emplaced during regional subhorizontal N-S contraction and E-W extension [*Gonzalez et al.*, 1996]. Strain accompanying pluton emplacement was facilitated by thermal softening of the host gneisses, resulting in an ~2-km-wide

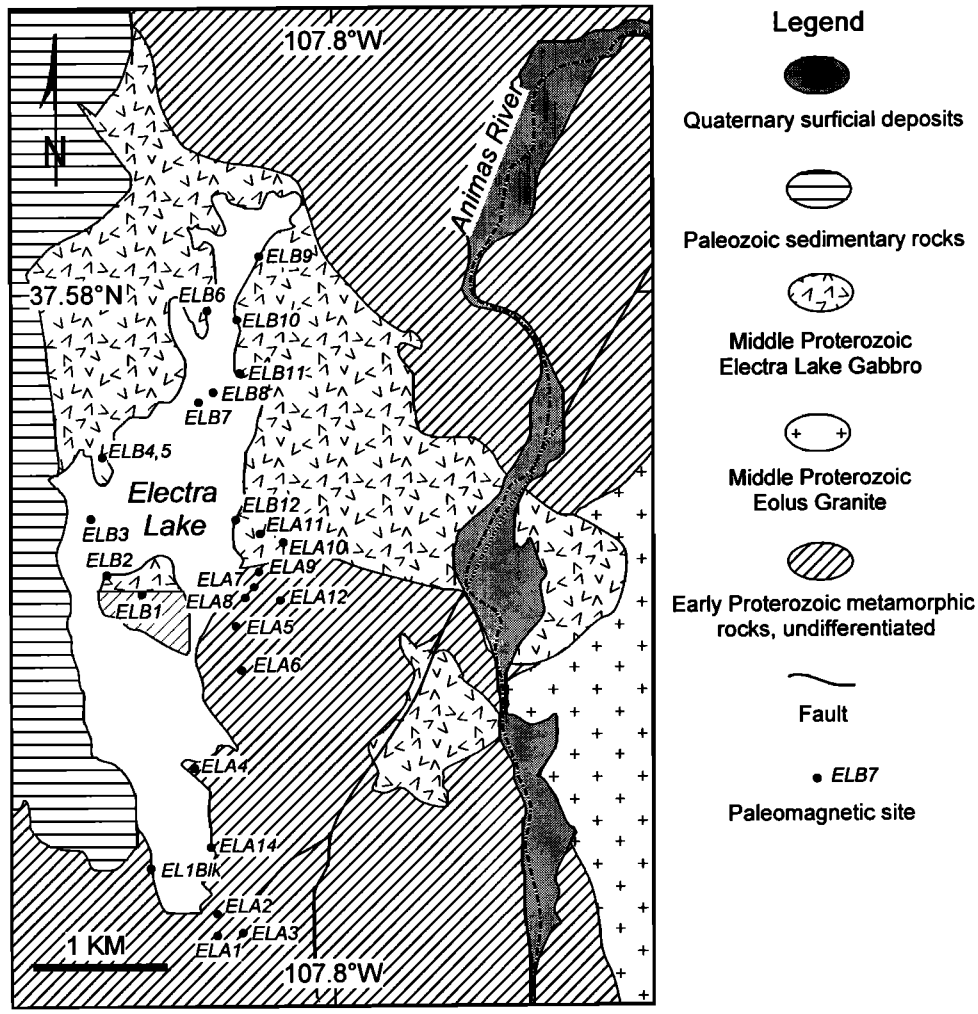


Figure 2. Generalized geologic map of the Electra Lake Gabbro, Needle Mountains, Colorado, showing the location of the paleomagnetic sample sites. Because of the generalized nature of the map, several small 1.4 Ga dikes and irregular intrusions that cut the Early Proterozoic host rocks have not been shown. Map modified from *Gonzales et al.* [1996].

thermal aureole along the southern margin of the ELG. From a more regional standpoint, magmatism and syncontractional deformation, including emplacement of the ELG, in the southwestern United States may be related to widespread crustal shortening along the southern margin of Laurentia between 1.5 and 1.3 Ga [*Nyman et al.*, 1994; *Kirby et al.*, 1995; *Gonzalez et al.*, 1996].

Paleomagnetic samples (4 to 16 samples/site) were collected from 25 sites around Electra Lake (Figure 2). Fifteen sites are from the ELG, primarily in fresh exposures along the shores of the lake and several small islands. Five sites were collected from mafic dikes that cut the Twilight gneiss. In order to evaluate the thermal effects of emplacement of the ELG on the nearby host rocks and to test for the possibility of regional thermomagnetic or thermochemical overprints, we collected samples from two sites in the melt zone adjacent to the southern margin of the ELG (sites ELA7 and ELA 8): one site in a gneiss screen in the gabbro along the western part of the exposed gabbro (site ELB4), one site in gneiss about 0.75 km from the southern gabbro/gneiss contact (ELA6), and one site in an amphibolite located about 2.5 km from the southern margin of the gabbro/gneiss contact (ELA2). At four of the dike sites, samples were collected in the host Twilight Gneiss in order to conduct baked contact tests [*Everitt and Clegg*, 1962]. With the exception of some melt zone samples at the southern margin of the ELG, most of the gneiss or amphibolite samples from the host basement rocks or from the baked contact tests with the dikes did not yield fully coherent or internally consistent paleomagnetic results, and they are not considered further.

3. Paleomagnetic Methods

Most paleomagnetic samples were obtained using a portable drill, with individual samples oriented by magnetic and solar compasses and a clinometer. Samples from two sites in visually sensitive localities were collected as oriented block samples using a Brunton compass. The natural remanent magnetization (NRM) of each specimen was measured using cryogenic or spinner magnetometers. Detailed progressive alternating field (AF) and/or thermal demagnetization was conducted on all samples. Demagnetization results were analyzed using orthogonal vector diagrams, stereographic pro-

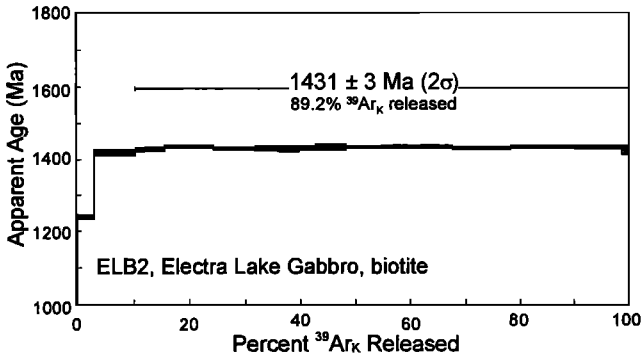


Figure 3. Biotite age spectrum from the Electra Lake Gabbro. The height of the rectangle for each temperature step represents the analytical error in the apparent age at ± 1σ. The line shows the steps used in the calculation of the weighted-mean plateau date. The error in the plateau date is reported at the 2σ level and includes the error in the flux parameter *J*.

jections, and normalized intensity decay plots. Characteristic and secondary components of magnetization, recognized by colinear demagnetization paths on vector diagrams, were calculated using principal components analysis [*Kirschvink*, 1980]. Where demagnetization failed to isolate a single characteristic direction or stable endpoint but yielded a curvilinear demagnetization trajectory due to overlapping coercivity and/or unblocking temperatures, we calculated demagnetization circles [*McFadden and McElhinny*, 1988]. Site-mean directions from sites consisting solely of well-defined linear data were calculated using statistics from *Fisher* [1953]; site-mean directions based on a combination of linear and great circle data were calculated using the maximum likelihood approach method of *McFadden and McElhinny* [1988]. Magnetic carriers were determined by demagnetization characteristics during AF and thermal demagnetization, acquisition and AF demagnetization of isothermal remanent magnetization (IRM), and anhysteretic remanent magnetization (ARM) demagnetization experiments and by Curie temperature determinations. Five Curie temperature determinations were made in air in saturating fields of 0.2-0.4 T using an electrobalance similar to that described by *Larson et al.* [1975]. Additionally, low-frequency magnetic susceptibility versus temperature thermomagnetic determinations on six samples were made in a flowing argon atmosphere using a Kappabridge similar to that described by *Hrouda* [1995].

4. Results

Paleomagnetic results from the ELG samples and samples from one of the melt zone sites generally fall into two types.

Table 1. $^{40}Ar/^{39}Ar$ Incremental Heating Data for Biotite From the Electra Lake Gabbro, Needle Mountains, Southwestern Colorado

Temperature, °C	$^{40}Ar_R$	$^{39}Ar_K$	$^{40}Ar_R/^{39}Ar_K$	$^{39}Ar/^{37}Ar$	$\%^{40}Ar_R$	$\%^{39}Ar$	Apparent Age, Ma ± 1σ
Sample ELB2, Biotite, ELG; 8.0 mg; measured $^{40}Ar/^{36}Ar_a$ = 298.9; J value = 0.007180 ± 0.1% (1σ).							
850	0.04186	0.00064	65.271	0.36	60.3	0.1	693±46
950	2.5101	0.01821	137.85	4.25	97.3	3.0	1241±5
1000	7.5737	0.04546	166.62	18.5	98.2	7.6	1419±7
(1025)	5.0703	0.03016	168.10	--	99.6	5.0	1428±2
(1050)	8.8890	0.05253	169.21	19.8	99.9	8.8	1435±2
(1100)	7.4835	0.04451	168.12	10.5	99.7	7.4	1428±2
(1133)	4.3729	0.02599	168.25	11.0	99.7	4.3	1429±8
(1167)	3.6206	0.02156	167.93	2.62	99.3	3.6	1427±7
(1200)	2.9949	0.01777	168.52	2861	99.2	3.0	1431±5
(1250)	5.5853	0.03300	169.27	8481	99.3	5.5	1435±9
(1300)	8.7680	0.05192	168.86	10.6	99.7	8.7	1433±2
(1333)	10.759	0.06367	168.98	62.3	99.8	10.6	1433±3
(1367)	10.867	0.06465	168.07	18.0	99.7	10.8	1428±3
(1400)	11.798	0.06983	168.96	23.9	99.9	11.7	1433±3
(1433)	8.5889	0.05097	168.52	15.2	99.9	8.5	1431±6
(1500)	1.3252	0.00791	167.43	4.10	99.2	1.3	1424±11
Total gas			167.42				1424±4

The $^{40}Ar_R$ is radiogenic ^{40}Ar in volts signal; $^{39}Ar_K$ is potassium-derived ^{39}Ar in volts signal; $^{40}Ar_R/^{39}Ar_K$ is the ratio of $^{40}Ar_R$ to $^{39}Ar_K$ after correction for mass discrimination and interfering isotopes; $^{39}Ar/^{37}Ar$ is the ratio of $^{39}Ar_K$ to $^{37}Ar_{Ca}$ (this value can be converted to the approximate K/Ca by multiplying by 0.52); $\%^{40}Ar_R$ and $\%^{39}Ar$ are the percent of radiogenic ^{40}Ar and percent of total ^{39}Ar released in each temperature step. Temperature steps in parentheses are those used in the calculation of the plateau age. Conversion of volts signal to moles Ar can be made using a conversion factor of 1.252×10^{-12} moles argon per volt of signal. The biotite used in $^{40}Ar/^{39}Ar$ dating was prepared from paleomagnetic samples from site ELB2 in the Electra Lake Gabbro. The sample was crushed and sieved and the mineral separates were concentrated using standard density and magnetic techniques; an optically pure (>99.9%) separate was produced by final hand-picking. The sample and flux monitors were loaded in aluminum foil capsules, placed in quartz vials, and irradiated in the central thimble of the U.S. Geological Survey TRIGA reactor. Vertical and horizontal variations in neutron flux in the irradiated package were monitored by 8 to 12 standards distributed along the length of each vial; the geometry of the package was such that each unknown sample was adjacent to at least one standard. The flux monitor used was hornblende MMHb-1, which has an assigned K-Ar age of 520.4 ± 1.7 Ma [*Samson and Alexander*, 1987]. Corrections for reactor-produced interfering reactions were made using argon isotopes of K_2SO_4 and CaF_2 irradiated in each package. Following irradiation, the biotite was progressively degassed in a double-vacuum resistance furnace for 20 min in 10 steps from 950° to 1550°C. Five naturally occurring and radiogenic argon isotopes (^{40}Ar, ^{39}Ar, ^{38}Ar, ^{37}Ar, and ^{36}Ar) were measured at each step using a Mass Analyzer Products MAP-215 mass spectrometer operated in the static mode. Apparent ages for each temperature step were calculated using the decay constants recommended by *Steiger and Jäger* [1977]. The determination of whether the apparent ages in the age spectrum constituted a plateau was made using the criteria of *Fleck et al.* [1977]. Following this criteria, a plateau is defined as comprising two or more contiguous gas fractions that yield apparent ages that are statistically identical at 95% confidence (using the critical value test of *Dalrymple and Lanphere* [1969]) and which together constitute greater than 50% of the total potassium-derived ^{39}Ar released in the step-heating experiment. The plateau age was calculated using a weighted mean, where weighting is by the inverse of the analytical variance [*Taylor*, 1982]. Calculation of the total gas and plateau dates includes the analytical uncertainty in the determination of the flux parameter *J*.

The first is characterized by samples that consist of essentially a single magnetization of northeast declination and moderate negative inclination. These samples show well-defined decay of the magnetization vector to the origin of vector diagrams during AF or thermal demagnetization (Figures 4a and 4c). This northeast declination, negative inclination magnetization (Figure 5a) is unlike that of the present day field or of any expected direction for the Phanerozoic. We consider this magnetization to be the characteristic remanent magnetization (ChRM) of the ELG.

The second type of demagnetization behavior is characterized by samples that contain two or more magnetizations. The first component removed during demagnetization is usually of north declination and moderate positive inclination, or is randomly directed (Figure 4b and 4d). These magnetizations are typically removed at low AF levels or laboratory unblocking temperatures. At higher AF fields or temperatures, a consistent magnetization of northeast declination and moderate negative inclination that decays toward the origin of demagnetization diagrams is isolated. Because of their low coercivities and laboratory unblocking temperature, the secondary magnetizations are probably viscous remanent magnetizations acquired in the present-day field or during laboratory storage that are superimposed upon the ChRM of the gabbro samples.

The AF and thermal demagnetization behavior and rock magnetic characteristics of the ELG samples indicate the ChRM is carried by nearly pure magnetite. AF demagnetization results indicate that the ChRM is carried by a low to

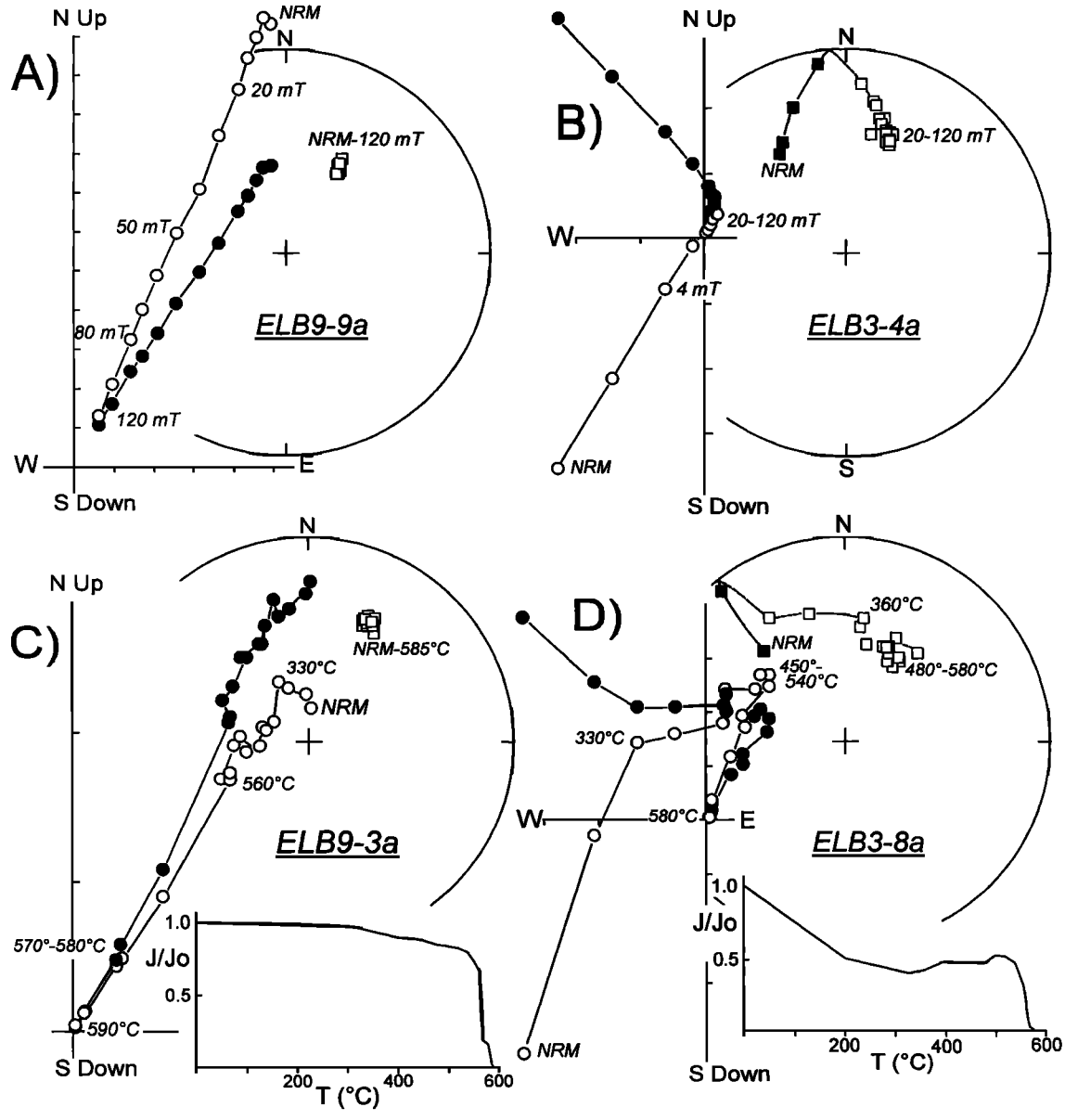


Figure 4. Representative orthogonal vector diagrams and equal-area projections showing behavior of samples of the Electra Lake Gabbro during alternating field (Figures 4a and 4b) and thermal demagnetization (Figures 4c and 4d). Also shown are normalized intensity versus temperature diagrams for thermally demagnetized samples. Open and solid symbols for the vector diagrams represent the in situ magnetization components projected onto the vertical and horizontal planes, respectively. For each vector diagram, one tic = 0.1 A/m. For the equal area projections, the open symbols are projections on the upper hemisphere and denote negative inclinations.

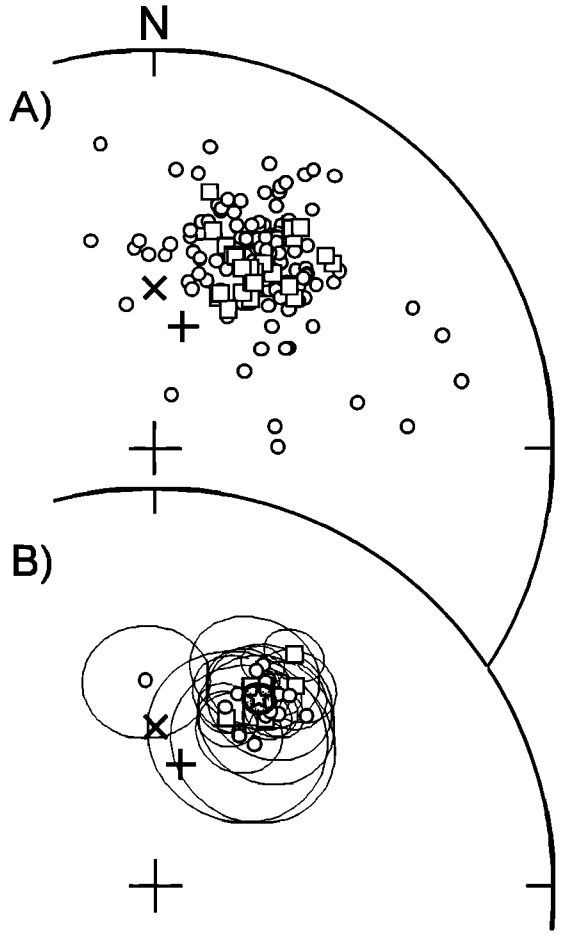


Figure 5. Equal-area projections showing in situ paleomagnetic results from the Electra Lake Gabbro. (a) Characteristic directions obtained from the Electra Lake Gabbro and mafic dikes. Circles are directions from the gabbro, squares are from mafic dikes. (b) Site-mean directions for the gabbro and dikes and their α_{95} cones of confidence. The grand-mean direction and its α_{95} are denoted by the star and heavy line. Open symbols are projections on the upper hemisphere and denote negative inclinations. The pluses and crosses are directions of the positive inclination, present-day and time-averaged magnetic fields for the study area are plotted on the lower hemisphere.

moderate coercivity phase, as median destructive fields (MDF) are typically between 10 to 60 mT (Figures 4a-4d and Figures 6a and 6b). Saturation of isothermal remanent magnetization (IRM) occurs in fields less than 200 to 300 mT (Figures 6a and 6b) also indicating that the magnetization is dominated by a low-coercivity phase, such as magnetite. Comparison of AF demagnetization of the induced IRM and anhysteretic remanent (ARM) magnetization in a modified *Lowrie and Fuller* [1971] test demonstrates that the weak field ARM is more resistant to demagnetization than the IRM (Figures 6a and 6b), a result consistent with the ChRM being carried by a mixture of grains of pseudosingle or single domain character. Normalized thermal decay curves indicate that the demagnetization of the ChRM occurs over a limited range (i.e., $\leq$ 40°C) of temperatures and that complete unblocking occurs by about 585°C (Figures 4c and 4d). Thermomagnetic analyses of magnetic separates from gabbro and dike samples using a Curie balance yields nearly reversible heating and cooling curves (Figure 7a) with a mean Curie temperature of 582° ± 5°C (one standard deviation, N = 5 determinations). Similarly, low-frequency magnetic susceptibility versus temperature curves also show nearly reversible curves (Figure 7b), with broad humped-shape region at about 300°C during heating, and pronounced upward inflections just below the "Curie" temperature attributable to the Hopkinson effect. The observed curves appear to be a combination of the idealized H- and single domain (SD)-type curves of *Radhakrishnamurty and Likhite* [1970]. These thermomagnetic determinations yield slightly lower, but statistically indistinguishable "Curie" temperature 576° ± 5°C (N = 6 determinations) than those obtained with the Curie balance. The results from both types of thermomagnetic determinations are consistent with the presence of nearly pure magnetite. The combination of relatively high MDFs, the results of the modified Lowrie-Fuller tests, the square-shouldered unblocking temperature curves, and the thermomagnetic determinations indicate that the ChRM is carried by nearly pure fine-grained

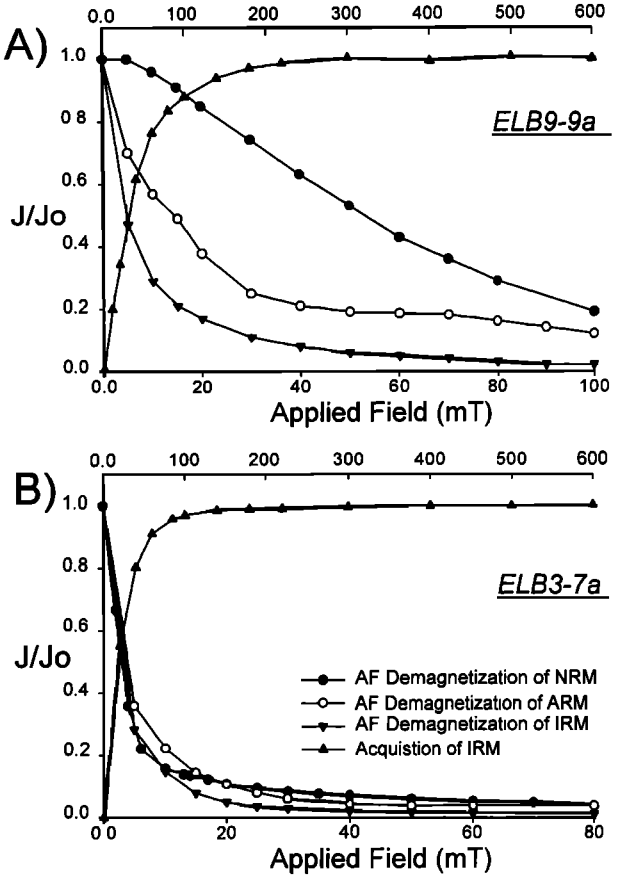


Figure 6. Representative diagrams showing the acquisition of an induced isothermal remanent magnetization (IRM) (upward pointing solid triangles, upper scale, both diagrams) and the AF demagnetization of natural and artificially induced magnetizations. The NRM results are shown by solid circles, the ARM by open circles, and the IRM by downward pointing solid triangles; all AF demagnetization results are referenced to the lower scale. (a) gabbro sample consisting of an essentially univectorial example of the ChRM that is carried by pseudosingle or single domain magnetite. (b) gabbro sample consisting of a mixture of grains of multidomain to pseudosingle/single domain character.

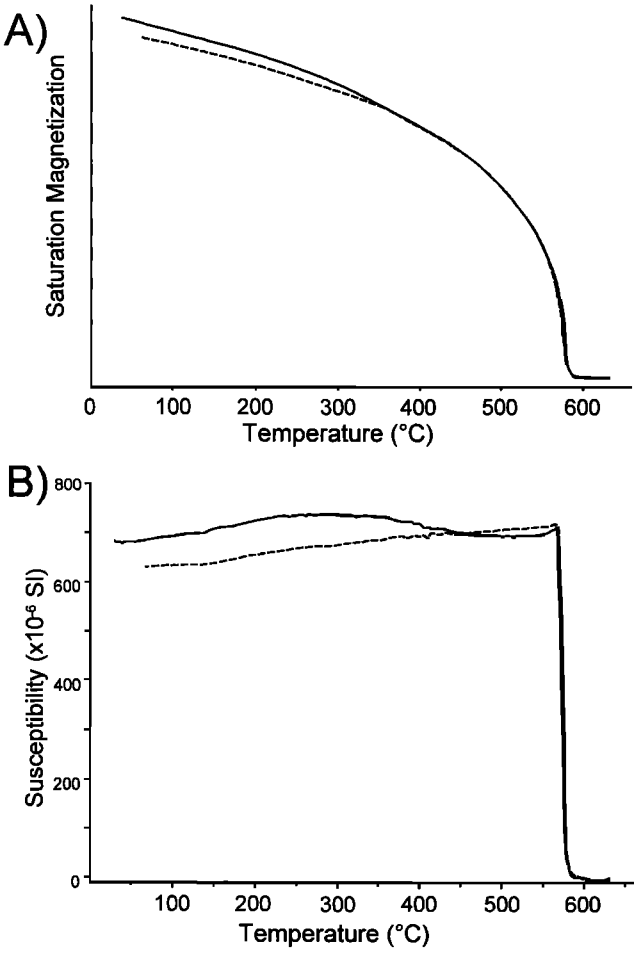


Figure 7. Thermomagnetic determinations of samples from the Electra Lake Gabbro. (a) Magnetization versus temperature determination made in air in a saturating field of 0.4 T. (b) Low-frequency magnetic susceptibility versus temperature determination made in a flowing argon environment. For both diagrams, solid and dashed lines refer to heating and cooling curves, respectively.

magnetite of single- or pseudosingle domain form. Secondary magnetizations of viscous origin probably reside in coarser multidomain magnetite. Although sulfides are relatively common in these gabbros, there is no evidence that paleomagnetically significant magnetizations are carried by these phases.

Twenty-one sites from the ELG and associated intrusive and melt zone rocks gave usable paleomagnetic results for calculation of site-mean directions. The mean directions are generally well-grouped at the site level, with 86% having α_{95} values of less than 15° (Table 2). A grand mean calculated from the 21 sites gives an in situ mean direction of declination (D) = 29.1° and inclination (I) = -45.7° (k = 94, α_{95} = 3.3°) (Figure 5B). Although layering in the ELG dips steeply, we cannot use it to correct our paleomagnetic data. Such corrections assume that the gabbro layering formed by crystal settling and is an accurate record of paleohorizontal. Numerous studies, however, show that igneous layering can form in virtually any orientation including overturned [*Rossman*, 1963; *Campbell*, 1978; *McBirney and Noyes*, 1979]. In some cases, folding of igneous layering in intrusive complexes predates acquisition of primary thermoremanent magnetizations [*Evans*, 1968; *Campbell*, 1978; *Harlan et al.*, 1994; *Gromme*, 1997]. However, we corrected the ELG group mean direction for the slight west directed tilt of the overlying Middle Cambrian Ignacio Formation strata (250°W, dip of 5°), giving a mean of D = 32.1° and I = -41.9°, which we consider the best directional estimate for the ELG.

5. Discussion

The ELG yields a well-defined ChRM that is probably a thermoremanent magnetization acquired at about 1435 Ma. The concordance between the magnetization direction in the gabbro and the melt zone in the contact rocks is consistent with a thermoremanent origin, although we cannot demonstrate that the ELG ChRM has been superimposed on a preexisting stable magnetization in the host basement rocks. Assuming that each accepted site-mean direction from the gabbro, dikes, and melt zone is an independent estimate of the geomagnetic field, we calculate a grand mean pole from the 21 tilt-corrected virtual geomagnetic poles (VGPs) (Table 2) that is located at 21.1°S, 221.1°E (K = 89, A_{95} = 3.4°) (Table 3). The estimated angular standard deviation for the VGPs, 8.6° ± 1.9°, is significantly less than the expected value of 14.5° ± 1.0° based on Model G of *McFadden et al.* [1991] for the interval 5-0 Ma. The low scatter suggests that the ELG has not sampled enough time to fully average paleosecular variation (PSV). This is a plausible interpretation given that the pluton is relatively small and probably cooled quickly, consistent with the overlapping U-Pb zircon and $^{40}Ar/^{39}Ar$ biotite dates from the ELG. Because the ChRM is carried by fine-grained magnetite with a limited range of laboratory unblocking temperatures, the remanence could have been acquired during a very limited span of time. Essentially instantaneous to rapid cooling of the pluton through the range of blocking temperatures required for magnetization acquisition could explain why the ChRM in the ELG is only of one polarity. Alternatively, the low directional scatter could also reflect smoothing of the observed directional dispersion during cooling. The observed angular dispersion of paleomagnetic directions from plutonic rocks is commonly less than expected values [*Dodson et al.*, 1978; *Frei et al.*, 1984; *Harlan et al.*, 1994], even where dual-polarity magnetizations are present. We caution that estimates of the angular dispersion of plutonic rocks alone may not give accurate assessments of whether paleosecular variation has been averaged. Given the range of rock types present in the ELG, the presence of an extensive thermal aureole along the southern margin of the complex, and evidence for significant magmatic differentiation, it is likely that the ELG has averaged PSV and thus may be a reliable estimate of the geomagnetic field at 1435 Ma.

Paleomagnetic results from the ELG have important implications for the circa 1.5-1.4 Ga part of the Middle Proterozoic North American APW path. Poles from circa 1.47 to 1.42 Ga intrusive rocks are tightly clustered at equatorial latitudes, with a mean at 0.5°S, 214.7°E (K = 153, A_{95} = 6.2°, N = 5 poles) (Table 3 and Figure 8). Poles from the Middle Proterozoic Belt Supergroup are typically 15° to 25° steeper in south latitude and somewhat west of those from the intrusions; a mean pole for the Belt Supergroup is located at 18.9°S,

Table 2. Paleomagnetic Data From the Electra Lake Gabbro, Needle Mountains, Southwestern Colorado

Site	Rock Type	Slat. °N	Slong. °W	N/N_o	Geographic		Stratigraphic		k	α_{95}, deg	Plong. °E	Plat. °S
					D, deg	I, inc	D, deg	I, inc				
EL1Blk	Pdb	37.5456	107.8064	6/6	25.3	-44.0	28.4	-40.3	69	8.1	223.6	23.8
ELA1	Pdb	37.5447	107.8028	4(1)/8	33.2	-48.0	36.2	-43.9	408	4.8	218.1	18.2
ELA3	Pdb	37.5444	107.8022	4/4	35.3	-38.5	37.3	-34.3	146	7.6	213.4	23.5
ELA4	Pdb	37.5547	107.8022	10(2)/10	23.2	-52.3	27.5	-48.7	70	5.9	227.3	18.0
ELA6	Pgn/Pdb	37.5619	107.7997	4/8	41.7	-42.8	43.7	-38.4	12	27.5	209.7	18.0
ELA7	Pgn mz	37.5650	107.7992	7(2)/8	34.7	-54.2	38.3	-50.0	15	16.4	219.2	12.8
ELA9	Pgb	37.5661	107.7989	6/10	25.1	-40.4	27.8	-36.8	31	12.2	223.0	26.3
ELA10	Pgb	37.5672	107.7978	5/6	33.3	-46.5	36.1	-42.4	212	5.3	217.6	19.2
ELA11	Pgb	37.5678	107.7981	12/12	29.0	-40.8	31.6	-37.0	457	2.0	219.6	24.7
ELA14	Pdb	37.5475	107.8019	10/12	31.1	-33.0	33.0	-29.1	62	5.6	215.6	28.5
ELB1	Pgb	37.5644	107.8021	6(2)/10	357.4	-47.3	2.4	-45.6	34	12.2	249.9	25.4
ELB2	Pgb	37.5661	107.8100	10(2)/10	31.9	-44.6	34.7	-40.6	17	12.2	218.1	21.1
ELB3	Pgb	37.5667	107.8092	10(2)/12	33.6	-48.3	36.6	-44.2	14	13.5	217.9	17.8
ELB5	Pgb	37.5714	107.8110	7(1)/8	35.1	-41.1	37.4	-36.9	75	7.1	214.4	22.0
ELB6	Pgb	37.5814	107.8022	12(4)/12	32.9	-42.3	35.4	-38.2	100	4.4	216.6	22.2
ELB7	Pgb	37.5744	107.8019	11(6)/12	28.5	-53.8	32.6	-49.9	32	8.5	223.6	15.2
ELB8	Pgb	37.5739	107.8006	7(3)/10	28.9	-54.6	33.0	-50.7	13	17.8	223.5	14.4
ELB9	Pgb	37.5847	107.7997	9/9	23.1	-46.6	26.6	-43.1	93	8.4	226.1	22.6
ELB10	Pgb	37.5803	107.8006	9/9	25.9	-38.4	28.4	-34.7	157	4.1	221.8	27.3
ELB11	Pgb	37.5733	107.7997	12(1)/12	35.6	-45.8	38.2	-41.6	76	5.0	215.5	18.8
ELB12	Pgb	37.5683	107.8006	12(3)/13	21.6	-50.4	24.7	-46.9	69	5.3	228.2	20.0

Pgn is Proterozoic gneiss; Pgb is Middle Proterozoic ELG; Pdb is Middle Proterozoic diabase dike. Pgn/Pgb is a mixture of Proterozoic gneiss and dike samples from the southern contact of the ELG (only the dike samples yielded interpretable paleomagnetic results); Pgn mz are melt zone samples from the southern margin of the ELG; Slat. and Slong. are latitude and longitude of the paleomagnetic site; N/N_o is the number of direct observations of sample directions used in calculating site-mean directions/samples measured (the value in parentheses is the number of great circles where both linear and great circle data are used); D and I are declination and inclination of the situ site-mean direction; geographic and stratigraphic refer to the in situ and tilt-corrected coordinates, respectively; k is precision parameter [*Fisher*, 1953; *McFadden and McElhinny*, 1988]; α_{95} is semiangle of the cone of 95% confidence about the site mean; and Plong. and Plat. are latitude and longitude of the virtual geomagnetic pole (VGP) calculated from the tilt-corrected site-mean direction. The negative inclination site-means have been inverted to their positive inclination antipode prior to VGP calculation. Although shown in Figure 2, samples from sites ELA2, ELA5, ELA8, and ELB4 (all in host gneiss) and ELA12 (gabbro) were unstable during demagnetization and a characteristic site-mean direction from these sites could not be isolated.

207.2°E (K = 62, A_{95} = 5.6°, N = 12 poles) (Table 3 and Figure 8). This discrepancy has been attributed to APW, possibly due to plate motion, and/or uncertainties in the age of rocks of the Belt Supergroup and its characteristic magnetizations [*Harlan et al.*, 1994]. Although the age of the uppermost rocks of the Belt Supergroup is problematic, recent U-Pb dating of zircons from sills that syndepositionally intrude the lowermost portions of Belt Supergroup strata (Moyie sills: 1468 ± 2 Ma), a metabentonite in the uppermost Belt carbonate (Logan Pass bentonite: 1449 ± 10 Ma), and a stratigraphically higher rhyolite lava flow (upper part of the Purcell Lava: 1443 ± 5 Ma), demonstrates that most (approximately two thirds of the stratigraphic section) of the Belt Supergroup was deposited between 1.47 and 1.43 Ga [*Aleinikoff et al.*, 1996; *Anderson and Davis*, 1995], a time period within the range of isotopic dates from the intrusions that have previously provided the highest-quality paleomagnetic data for the Middle Proterozoic. This suggests that other problems may exist regarding the reliability of the Middle Proterozoic poles. Such problems include, but are not limited to, unrecognized systematic tilting of Middle Proterozoic intrusions, uncertainties in the absolute age of the uppermost rocks of the Belt Supergroup, uncertainties in the age of remanence acquisition in both types of rocks, and the possibility of unrecognized vertical axis rotations in both intrusions and Belt Supergroup rocks. Because the ELG is overlain in part by Paleozoic strata, paleomagnetic data from the gabbro can be corrected for at least the effects of Phanerozoic tilting and could be useful in refining the 1.5-1.4 Ga part of the North American APW path.

Comparison of the ELG pole with other 1.5-1.4 Ga poles (Figure 8) shows that it is located 30° to 35° south and somewhat east of the intrusions from elsewhere in North America and 31° from the mean intrusive pole. The ELG pole is similar in latitude to, but plots east of, most poles from the Middle Proterozoic Belt Supergroup and 13° from the mean Belt pole. Assuming that the ELG provides a reliable recording of the geomagnetic field at 1.43 Ga, the discrepancy between poles from the ELG and other intrusions of similar age, and the near concordance with those of the Belt Supergroup may imply that important problems exist regarding the reliability of the other 1.4 Ga igneous poles. Convincing paleomagnetic field tests to demonstrate the stability of remanent magnetizations in the Belt Supergroup rocks have not been described, but these strata reportedly contain a coherent magnetostratigraphy, which suggests that their magnetizations are primary or were acquired shortly after deposition [*Elston and Bressler*, 1980; *Link et al.*, 1993]. The discordance between the ELG , Belt Supergroup, and other 1.4 Ga igneous poles could also result from unrecognized systematic tilting and/or vertical axis rotations of the other 1.4 Ga intrusions, such that their poles plot at shallower latitudes. This seems unlikely as the rocks that provide these data are located from Wyoming to Labrador, and such an explanation is difficult to accept as it would require essentially uniform sense and magnitude of tilting over such a broad region. Paleomagnetic data from intrusions with no record of paleohorizontal must always be considered of lesser reliability than those from stratified rocks. However, the excellent agreement between poles from rocks greater than 2000 km apart strongly suggests that

Table 3. Middle Proterozoic Poles From North America for the 1.5 to 1.4 Ga Interval

Rock Unit and Location	Symbol (Figure 8)	Plat.	Plong.	$\delta p/\delta m$ or A_{95}	Age, Ma	Reference
Middle Proterozoic Intrusions						
Michikamau anorthosite	MK	1.5°S	218.0°E	4.5°	1460 ± 5	*Murthy et al.* [1968] and *Emslie et al.* [1976]
Croker Island Complex	CI	5.4°N	217.2°E	6.7°/10.5°	1475 ± 50	*Palmer* [1969]
Harp Lake anorthosite	HL	1.6°N	206.3°E	4.4°	1450 ± 5	*Irving et al.* [1977]
St. Francois volcanics	SF	1.1°S	216.9°E	5.2°/8.1°	1473 ± 2	*Hays and Scharon* [1966] and *Hsu et al.* [1966]
Laramie anorthosite complex/Sherman Granite	LAC/SG	6.7°S	215.0°E	3.5°	1440-1421	*Harlan et al.* [1994]
Mean Pole, Intrusions		0.5°S	214.7°E	6.2°		
Electra Lake Gabbro	ELG	21.1°S	221.1°E	3.4°	1435-1431	this study
Belt Supergroup Poles						
Pilcher Quartzite	B1	17.7°S	214.0°E	4.0°/6.9°	<1443	*Elston and Bressler* [1980]
McNamara Formation	B2	12.2°S	203.2°E	3.2°/5.4°	<1443	*Elston and Bressler* [1980]
Bonner Quartzite	B3	9.0°S	204.7°E	2.7°/4.5°	<1443	*Elston and Bressler* [1980]
Striped Peak Formation (Unit 4)	B4	8.5°S	195.9°E	2.3°/4.0°	<1443	*Elston and Bressler* [1980]
Striped Peak Formation (Unit 1)	B5	19.1°S	196.2°E	3.3°/6.2°	<1443	*Elston and Bressler* [1980]
Mount Shields Formation	B6	22.8°S	211.6°E	3.0°/5.6°	<1443	*Elston and Bressler* [1980]
Wallace Formation	B7	24.2°S	200.0°E	3.4°/6.6°	<1443	*Elston and Bressler* [1980]
Shepard Formation	B8	24.8°S	213.2°E	4.2°/7.9°	<1443	*Elston and Bressler* [1980]
Snowslip Formation	B9	25.1°S	204.4°E	2.1°/4.1°	1447-1443	*Elston and Bressler* [1980]
Empire Formation	B10	19.1°S	205.6°E	4.0°/7.4°	1447-1443	*Elston and Bressler* [1980]
Spokane Formation	B11	27.2°S	212.9°E	2.8°/5.4°	1447-1443	*Elston and Bressler* [1980]
Spokane Formation	B12	15.5°S	225.5°E	3.7°/6.2°	1447-1443	*Vitorello and Van der Voo* [1977]
Mean Pole, Belt Supergroup		18.9°S	207.2°E	5.6°		

Plat. and Plong. are the latitude and longitude of the paleomagnetic pole; $\delta p/\delta m$ are the minor and major semiaxes of the 95% confidence region about the pole, where calculated in direction space; A_{95} is the 95% confidence region about the pole, where calculated in pole space. Although recent contributions [e.g., *Link et al.*, 1993] discuss the implications of recent Belt Supergroup paleomagnetic data, most Belt Supergroup poles used in this compilation are from *Elston and Bressler* [1980] because it is the only reference that lists sufficient information (i.e., pole positions, statistics, sampling locations, etc.) for Belt Supergroup data to permit meaningful paleomagnetic interpretations. We have calculated a new pole for the St. Francois volcanics from the data listed by the referenced sources after excluding site-mean directions with N < 3 and $\alpha_{95} > 20°$. This gives an in situ mean direction of $D = 244.2°$, $I = 46.8°$ ($k = 78$, $\alpha_{95} = 6.3°$, $N = 8$ sites) and the pole for the St. Francois volcanics listed. Ages for the poles are from the following sources: Michikamau anorthosite and Harp Lake Complex are U-Pb zircon dates from the Global Paleomagnetic Database; the age for the Croker Island Complex is a Rb-Sr age whole rock isochron [*Palmer*, 1969]; the apparent age for the St. Francois volcanics pole is based on a weighted mean of 12 U-Pb determinations (excluding the Munger Granite Porphyry result) listed in Table 8 of *Van Schmus et al.* [1993]; the age range for the combined Laramie anorthosite complex/Sherman Granite pole is based on the range of published U-Pb dates from these units and a $^{40}Ar/^{39}Ar$ hornblende date as described by *Harlan et al.* [1994]; the age range for the ELG is based on the U-Pb zircon date for a pegmatitic gabbro that cuts the ELG and the $^{40}Ar/^{39}Ar$ biotite date described in this report; and ages for the Belt Supergroup poles are from the range of U-Pb determinations for sills that intrude the lower part of the Belt Supergroup and from volcanic rocks in the upper part of the Belt Supergroup stratigraphic section [*Anderson and Davis*, 1995; *Aleinikoff et al.*, 1996].

postemplacement tilts and rotations must have been minimal and probably random in nature and that poles from the igneous rocks are reliable recordings of the geomagnetic field at 1.47-1.43 Ga.

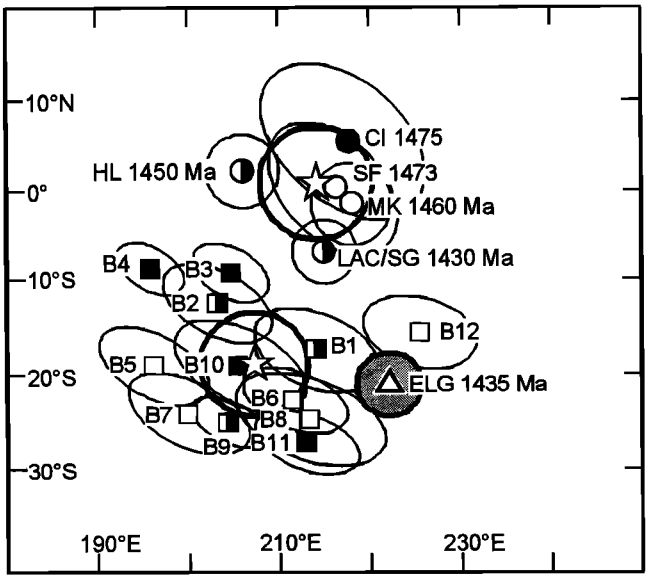


Figure 8. Miller projection showing 1.5 to 1.4 Ga paleomagnetic poles from North America and their 95% confidence limits. The triangle shows the location of the tilt-corrected Electra Lake Gabbro pole (ELG); circles are from 1.5 to 1.4 Ga intrusions; squares are poles from the Middle Proterozoic Belt Supergroup. Open symbols are poles from rocks with exclusively reverse polarity magnetizations (present-day coordinates); solid symbols are from rocks with normal polarity magnetizations; and half-solid symbols are from rocks which contain dual-polarity magnetizations. Stars show the mean poles for the 1.5 to 1.4 Ga intrusions and Belt Supergroup, as discussed in the text. Sources for the paleomagnetic poles and the symbols used are listed in Table 3.

A significant question also exists regarding the effects of vertical axis rotations on the paleomagnetic data from the Belt Supergroup. Most of the rocks from which the Belt poles are derived are allochthonous, having been thrust east during the Late Cretaceous to early Tertiary. Several studies demonstrate that parts of the thrust belt rotated both clockwise and counterclockwise during Mesozoic contraction and/or early Cenozoic extension [*Elston and Bressler.*, 1980; *Eldredge and Van der Voo*, 1988; *Jolly and Sheriff*, 1992; *Price and Sears*, 1996]. At present, it is difficult to assess the extent to which

such local rotations affect the Belt Supergroup data or what the implications of such rotations may have for the reliability of the Belt Supergroup paleomagnetic data with respect to the North American APW path.

As noted above, the latitude of the ELG pole is more consistent with poles of the Belt Supergroup than those from other 1.4 Ga intrusions. As a final consideration in this comparison, however, we return to the assumption that correction for tilt of Paleozoic strata has effectively corrected the ELG pole for postemplacement deformation. Synkinematic emplacement of the ELG and Eolus Granite during Middle Proterozoic N-S contraction [*Gonzalez et al.*, 1994] could have involved postemplacement, subsolidus tilt and/or rotation during the latter stages of Middle Proterozoic deformation. If so, this unrecognized deformation could account for some of the discordance between the ELG and other 1.5-1.4 Ga poles. For example, 25° of tilt about an axis trending 60° and plunging 40°NE would be required to bring the ELG pole into agreement with the mean 1.47-1.43 Ga intrusive pole, whereas 17° of tilt about an inclined axis of 34° with a declination of 342° would bring the ELG pole into agreement with the mean Belt Supergroup pole. In practice, however, it is impossible to demonstrate that tilting about these axes or any other possible axis occurred, let alone perform meaningful corrections to account for such deformation.

6. Conclusions

The 1435 Ma ELG has a well-defined thermoremanent magnetization of northeast declination and moderate negative inclination that is carried by magnetite. A paleomagnetic pole from the ELG and related dikes is slightly to significantly discordant with respect to poles from other time-equivalent rocks from North America. The reasons for this discrepancy are not clear, but could reflect a number of factors including failure to fully average PSV, unrecognized tilting and/or rotation of the complex, or structural and paleomagnetic complications inherent in poles used to define the 1.5-1.4 Ga part of the North American APW path. Additional high-quality poles are necessary in order to more precisely define the APW path for this important interval of time.

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